AI-Powered Robust Interaction Force Control of a Cardiac Ultrasound Robotic System

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***Abstract*—This article introduces a novel intelligent robust interaction force control method for a cardiac ultrasound robotic system (CURS), exploiting dual control loops and artificial intelligence (AI)-driven image feedback to enhance both image quality and patient safety during cardiac examinations. Unlike existing systems that use a constant interaction force, the proposed method adjusts the force based on ultrasound image feedback, which is critical for adapting to different cardiac views. The system employs an internal control loop, where the force feedback generates control commands (low-level controller), and an external control loop, where the feedback is processed through a convolutional neural network (CNN), named ultrasound-cardiac-feature-net (UCF-Net), determines the optimal force values (high-level controller). An adaptive filtered quasi-sliding mode controller (AFQSMC) manages both interaction force and probe's position within a hybrid position/force control context, ensuring robustness against uncertainties and disturbances. Experimental evaluations on a cardiac phantom navigating main cardiac views demonstrate the superiority of the proposed approach over traditional constant force control. Moreover, AFQSMC achieves significant improvements in interaction force control, with enhancements ranging from 21.87% to 68.25% over traditional FQSMC, sliding mode control (SMC), and proportional-integral (PI) controllers, across quantitative metrics such as root mean square (RMS), standard deviation (STD), and Max, confirming its potential for improving cardiac examination performance.**

***Index Terms*—Adaptive filtered quasi-sliding mode controller (AFQSMC), cardiac ultrasound robotic system (CURS), interaction force control, ultrasound-cardiac-feature-net (UCF-Net).**

I. Introduction

CARDIOVASCULAR disease is recognized as one of the main causes of death worldwide [1]. Early diagnosis and intervention through examination are critical to limit the risks of mortality and morbidity in patients with cardiovascular disease. Among various examination methods, ultrasound scanning is preferred due to its nonionizing nature, portability, and cost-effectiveness [2]. However, access to ultrasound devices and expert sonographers remains challenging in remote and underprivileged areas, such as rural regions.

To address the aforementioned issue, researchers have proposed the development of ultrasound robots capable of performing remote or automatic examinations [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. In [4], a six-DOF robotic arm equipped with an ultrasound probe was introduced for remote ultrasound examinations operated by an expert. This setup involved an ultrasound-like probe, which allowed the doctor to manipulate the probe and transmit its desired pose from their end to the patient's end via the Internet. The ultrasound image results were returned to the doctor in real time, enabling them to scan the patient's body. In [5], a similar tele-operated robotic ultrasound system was developed, which included a Universal Robot UR5. A haptic technique was also integrated for force feedback at the doctor's end, allowing more realistic and accurate operations. In [6], an ultrasound robot was designed to reconstruct 3-D ultrasound images from 2-D images. For this purpose, the robot was equipped with a depth camera to estimate the surface of the patient's body, and the desired path on the body's surface was tracked to scan the internal organs. In [7], an ultrasound robot was developed for breast scanning. Initially, the breast was scanned using a depth camera, and then the ultrasound scan interaction was initiated. Note that among the different types of ultrasound robotic systems, the development of a cardiac ultrasound robotic system (CURS) to specifically perform cardiovascular examination has been rare. Recently, in [8], a CURS was introduced for the first time for remote diagnostic purposes, allowing a cardiologist to examine patients located in underprivileged areas. However, exploiting robust and intelligent control methods was not reported.

One key factor for ultrasound robots is the control of interaction force [3]. On one hand, to ensure safety, the interaction

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force between the ultrasound probe and the patient's body must be minimized. On the other hand, a specific amount of force is necessary for full compliance and high-quality ultrasound image acquisition, depending on the applications. Some studies have considered interaction force control in ultrasound robotic systems. In [4], a six-axis force/torque sensor was attached to the robot's end-effector, ahead of the ultrasound probe, to monitor the interaction force. However, interaction force control was not implemented. In [5], the interaction force was controlled to ensure both safety and full compliance between the ultrasound probe and the body, using a hybrid position/force controller (HPFC). A proportional integral (PI) controller was employed for the force control task. In [6], impedance control was adopted to manage the interaction force. However, impedance control does not directly regulate the interaction force to a specific value, which can lead to force variations during movement over uneven surfaces on the patient's body. In [7], force control was considered not only for safe interaction but also for estimating the orientation of the ultrasound probe. The objective was to maintain the ultrasound probe's orientation perpendicular to the surface to improve ultrasound image quality during breast examinations. In [9], a flexible probe holder was introduced, facilitating interaction force control on stiffer bone surfaces. It also acted as an energy buffer, thereby increasing safety. A similar mechanism was designed in [11] for spine 3-D reconstruction. In [12], an inverse reinforcement learning method (IRL) was employed for full compliance force control based on two control loops: a position control loop and a posture control loop. IRL was utilized for both posture and force control.

In the reviewed literature, the feedback from force sensors has been primarily utilized for interaction force control to ensure safety during ultrasound examinations. However, existing methods often overlook the potential of ultrasound image feedback in adjusting the desired interaction force, which is crucial for obtaining high-quality images during cardiac examinations. These examinations require the probe to maintain contact with the body at various poses and apply different force levels, filling a significant gap in the current technology. Moreover, the robustness of these methods against uncertainties, such as variable stiffness and damping coefficients of the interaction, as well as measurement noise from the force sensor, remains inadequately addressed.

Motivated by the shortcomings and inspired by the advantageous features identified in the literature, in this study, an intelligent interaction force control system is developed. This system defines the desired force value by leveraging ultrasound image feedback analyzed through an artificial intelligence (AI) model. Additionally, a novel deep adaptive robust interaction force control method has been developed to accurately track the desired force signal, even when the system is subject to model uncertainties and sensor noises. This approach not only addresses the highlighted limitations but also significantly enhances the adaptability and precision of force control during ultrasound cardiac examinations.

The proposed method includes an internal loop that implements hybrid position/force control (HPFC) based on an orthogonal sliding manifold (OSM). This approach allows for HPFC to use a single sliding mode controller (SMC) rather than two independent controllers [13]. SMC is part of a variable structure system (VSS) known for its robustness against large uncertainties [14]. However, conventional SMC suffers from the chattering issue, which emanates from a discontinuous term in the control law and may damage the physical components of the device, such as actuators and joints. Quasi-SMC (QSMC) is an effective and feasible approach that addresses the chattering issue by replacing the discontinuous term with a continuous one. In [13], a filtered QSMC (FQSMC) was proposed to filter out measurement noises thanks to an intrinsic filter within its control law. It is important to note that the chattering-free feature of QSMC comes at a cost, resulting in increased tracking error due to the thickness of the boundary layer which mostly depends on the uncertainty bounds [13], [14], [15]. To address this issue, this study develops an adaptive estimator/compensator to reduce the tracking error. Since the states of the system are the ultrasound image and the interacting force, a deep convolutional neural network (DCNN) based on VGG-16 with some modifications, named ultrasound cardiac-feature-net (USCF-Net), is developed to extract some unique deep image features. The USCF-Net is followed by radial basis function neural networks (RBFNN) leading to USCF-Net-RBF-1, which first estimates the uncertainty and then compensates for it. It results in the adaptive FQSMC (AFQSMC), which leads to decreasing the uncertainty bounds interval and thus achieving thinner boundary layer and higher accuracy.

Some investigations show that the interaction force can be in the range of 3 to 40 N for abdominal scanning [17]. However, this range directly depends on the ultrasound probe footprint, since it is the pressure that may harm the patients. Therefore, the admissible pressure range is investigated for cardiac examination in this study. Note that the desired interaction force for a CURS depends on the main cardiac views [16], and it may vary for each patient. For instance, the minimum required force for scanning the subcostal (SC) view is higher than that of parasternal long/short axis view, which might be different from one person to another. To address this issue, in this study, the desired force value is determined by an AI agent that uses UCF-Net followed by an RBFNN, named UCF-Net-RBF-2, placed in the external loop as the high-level controller. Thus, UCF-Net-RBF-2 takes the 2-D ultrasound image in real-time as feedback and outputs the desired interaction force. UCF-Net has been trained on a set of collected pairs of ultrasound images and ultrasound probe poses with respect to the patient's body/phantom. Overall, the main contributions of this article are summarized as follows.

1) Development of a cardiac ultrasound robotic system for intelligent and safe remote cardiac examinations.
2) Development of the UCF-Net to extract unique cardiac ultrasound image features to be used for an intelligent dual-loop interaction force controller.
3) Development of the UCF-Net-RBF-1 (based on UCF-Net) for adaptive estimator/compensator. The integration of this adaptive mechanism with FQSMC leads to AFQSMC, which is robust against uncertainties and noise with high tracking accuracy, serving as the low-level controller.
4) Development of the UCF-Net-RBF-2 (based on UCF-Net) to determine the desired force value during cardiac

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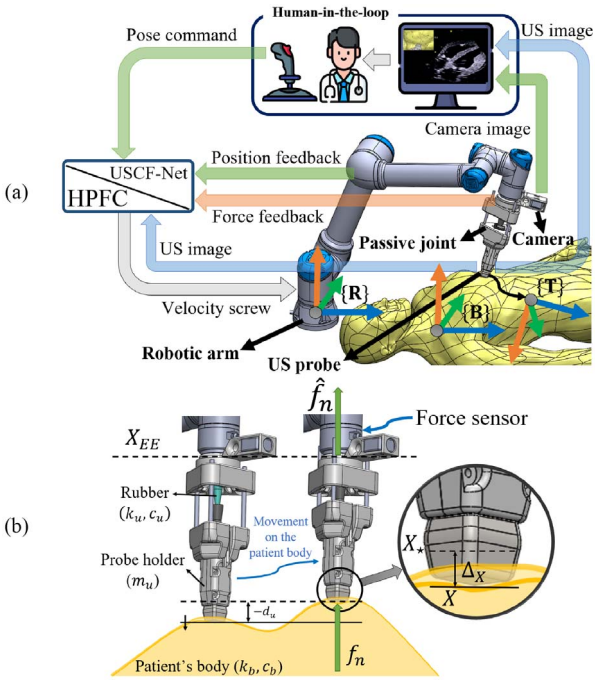


Fig. 1. (a) Schematic of a six-DOF US robot equipped with a force sensor interacting with a human body/phantom. (b) Passive joint and interaction model.

ultrasound examinations using ultrasound image feedback, serving as the high-level controller.

The rest of the article is organized as follows. Section II presents the problem statement and modeling of the CURS. The development of the UCF-Net is described in Section III. The proposed control method is developed in Section IV. Experimental results are presented in Section V. Conclusion remarks are drawn in Section VI.

II. CURS Modeling and Problem Statement

A. Problem Statement

The CURS consists of a ultrasound probe attached to the end-effector of a six-degree-of-freedom (six-DOF) robotic arm. The built-in force sensor is placed on the robotic arm's end-effector to measure the wrench. The ultrasound probe frame (tool frame) $\{\mathbf{T}\}$ is attached to the top of the ultrasound probe, and the reference frame $\{\mathbf{R}\}$ is fixed on the base of the robot, and a fixed frame is also attached to the surface of the patient body $\{\mathbf{B}\}$ as shown in Fig. 1(a). Besides, inspired by [9] and [11], a prismatic passive joint is used as a probe holder [Fig. 1(a)]. This joint, connecting robot parts with rubber, is built without an actuator, making it nonactive. It enhances safety by absorbing energy during possible faults and preventing force spikes during sudden interactions with the human body, making the ultrasound examination more comfortable and safer.

The main objective of this study is to solve the intelligent force control problem of the CURS when interacting with a patient while HPFC is applied. The intelligent component determines the desired force value based on ultrasound image feedback for different cardiac views, thereby eliminating the sonographer's need to adjust it during remote examinations.

One may define the HPFC problem as follows [18]:

$$\lim_{t\to\infty} \mathbf{Q}e_F = 0,\ \lim_{t\to\infty}((\mathbf{I}-\mathbf{Q})^T_R\mathrm{f}(e_X)) = 0 \tag{1}$$

where $e_X \in \mathbb{R}^6$ and $e_F \in \mathbb{R}^6$ are vectors of the robot's pose and wrench error as follows:

$$e_X = X - X_d \tag{2}$$

$$e_F = F - F_d \tag{3}$$

where $X \in \mathbb{R}^6$ is the ultrasound probe's pose attached to the robot's end-effector and $X_d \in \mathbb{R}^6$ is the desired pose. $F \in \mathbb{R}^6$ is the wrench and $F_d \in \mathbb{R}^6$ is the desired wrench which will be determined automatically by an AI agent. In (1), ${}^T_R\mathrm{f}$: $\mathbb{R}^6 \to \mathbb{R}^6$ is the rotation function that maps the pose error from $\{R\}$ to $\{T\}$. and $\mathbf{Q} \in \mathbb{R}^{6\times6}$ is the compliance selection matrix as follows:

$$\mathbf{Q} = DD^{\mathrm{T}} \tag{4}$$

where D in (4) is defined as follows:

$$D = \begin{bmatrix} D_P & D_R \end{bmatrix}^{\mathrm{T}},\ D_P = \begin{bmatrix} 0 & 0 & inx_c \end{bmatrix}^{\mathrm{T}}, D_R = 0_{3\times1} \tag{5}$$

where, $inx_c \in \{0,\ 1\}$ denotes the control mode. If the robot is in touch with the patient's body, it gets a value of 1; otherwise, it gets a value of 0. Note that since normal force is controlled during the interaction, only the third entry of D is inx_c.

B. Motion and Interaction Force Model

Developing an appropriate HPFC can realize the control objective (1). The input to the robotic system is the velocity screw $V \in \mathbb{R}^6$ of the robots' end-effector, i.e., $V = \dot{X}_{EE}$ and the output is considered the pose of the ultrasound probe X and interaction force/torque F applied to the probe's tip. The interaction normal force can be modeled assuming the body and the passive joint behave as spring and damper, as shown in Fig. 1(b). One may derive the normal interaction force relation as follows:

$$f_n = k_b D^{TT}_R\mathrm{f}(\Delta_X) + c_b D^{TT}_R\mathrm{f}(\dot{\Delta}_X) \tag{6}$$

where $f_n \in \mathbb{R}$ is the normal force applied to the ultrasound probe's tip meeting the following relation:

$$Df_n = DD^{\mathrm{T}}F = \mathbf{Q}F. \tag{7}$$

In (6), $k_b \in \mathbb{R}$, and $c_b \in \mathbb{R}$ are the stiffness and damping coefficients of the patient's body at the interaction point. $\Delta_X \in \mathbb{R}^6$ is the probe's pose displacement calculated as follows:

$$\Delta_X = X - X_\star \tag{8}$$

where $X_\star \in \mathbb{R}^6$ is the pose in which the probe touches the patient's body. The normal force measured by the force sensor is calculated as follows:

$$\hat{f}_n = k_u d_u + c_u \dot{d}_u \tag{9}$$

where $\hat{f}_n \in \mathbb{R}$ is the measured normal force, $k_u \in \mathbb{R}$, and $c_u \in \mathbb{R}$ are the stiffness and damping coefficients of the passive joint, respectively, and $d_u \in \mathbb{R}$ is the displacement of the passive joint

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[as shown in Fig. 1(b)]. Accordingly, one may derive the dynamic equation of the probe holder link as follows:

$$m_u D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\ddot{\Delta}_X\right) = -\widehat{f}_n + f_n + m_u D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\begin{bmatrix} 0 & 0 & \mathrm{g} & 0_{1\times 3} \end{bmatrix}^{\mathrm{T}}\right) \tag{10}$$

where $m_u \in \mathbb{R}$ is the mass of the probe holder. Assuming that the acceleration of the probe holder is neglectable (quasi-static), the left-hand side of (10) equals zero, which yields

$$f_n = \widehat{f}_n - m_u D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\begin{bmatrix} 0 & 0 & \mathrm{g} & 0_{1\times 3} \end{bmatrix}^{\mathrm{T}}\right). \tag{11}$$

Note that the relation above is used for indirect measurement of the interaction force applied to the ultrasound probe's tip. Considering uncertainties in the interaction model, i.e., $k_b = \widehat{k}_b + \delta_{kb}$, $c_b = \widehat{c}_b + \delta_{cb}$, (6) can be rewritten as follows:

$$f_n = D^{\mathrm{T}} F = \widehat{k}_b D^{\mathrm{T}}{}_R^T \mathrm{f}(\Delta_X) + \widehat{c}_b D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\dot{\Delta}_X\right) + \bar{\delta}_b \tag{12}$$

where notation $\widehat{\ }$ indicated the nominal value and $\bar{\delta}_b \in \mathbb{R}$, which denotes the uncertain part of the interaction model, is derived as follows:

$$\bar{\delta}_b = \delta_{kb} D^{\mathrm{T}}{}_R^T \mathrm{f}(\Delta_X) + \delta_{cb} D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\dot{\Delta}_X\right). \tag{13}$$

Taking time derivative of both sides of (12) yields

$$\dot{f}_n = D^{\mathrm{T}} \dot{F} = \widehat{k}_b D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\dot{\Delta}_X\right) + \widehat{c}_b D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\ddot{\Delta}_X\right) + \dot{\bar{\delta}}_b. \tag{14}$$

One may derive the end-effector displacement Δ_{XEE} as follows:

$$\Delta_{X_{EE}} = \Delta_X - {}_R^T \mathrm{f}^{-1}(D d_u) \rightarrow \Delta_X = \Delta_{X_{EE}} + {}_R^T \mathrm{f}^{-1}(D d_u). \tag{15}$$

By taking the first and second-order time derivatives of (15) and then substituting it into (14) the following relation yields:

$$\dot{f}_n = D^{\mathrm{T}}{}_R^T \mathrm{f}\left(\widehat{k}_b \dot{\Delta}_{X_{EE}} + \widehat{c}_b \ddot{\Delta}_{X_{EE}}\right) + D^{\mathrm{T}}\left(\widehat{k}_b \dot{d}_u + \widehat{c}_b \ddot{d}_u + \dot{\bar{\delta}}_b\right). \tag{16}$$

III. USCF-Net

A. UCF-Net Architecture

To develop an adaptive compensator and an external force controller, a DCNN, named UCF-Net is developed to take ultrasound images and return some unique deep image features. UCF-Net is a DCNN designed based on the VGG16 model [19], except it takes a grayscale image (ultrasound-Image) and returns a vector of six real numbers as the deep ultrasound image feature (DUIF) (a regression DCNN model). The network architecture of UCF-Net is depicted in Fig. 2(a). Referring to this figure, Ultrasound -Feature-Net has two main parts: decoder and regressor. The decoder consists of thirteen convolutional layers followed by pooling layers to decrease the dimension of the tensor size and rectified linear unit (ReLU) activation functions. The receptive fields are of size 3×3. The input, which takes a grayscale image is $\mathbf{I_C} \in \mathbb{R}^{224\times 224}$. The ultrasound image in B-mode goes through preprocessing. The preprocessing includes converting the image from RGB to grayscale, masking the intended area of the image to remove the unwanted areas, max-min normalization, and finally, resizing. The preprocessing stages are shown in Fig. 3. The regressor part comprises three fully connected layers

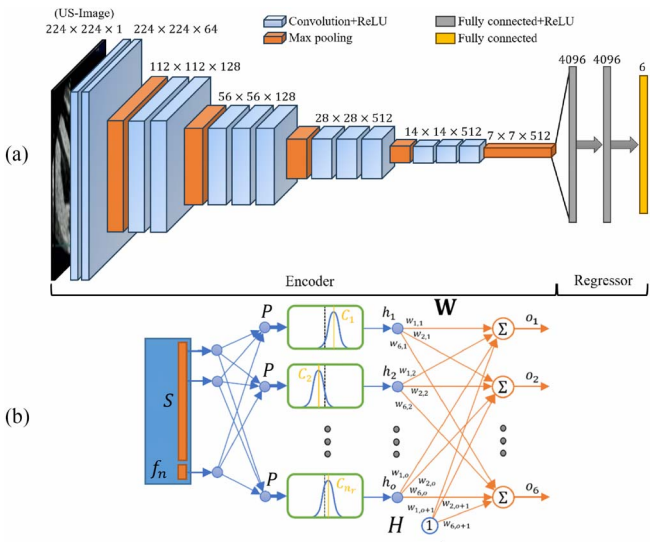


Fig. 2. (a) UCF-net architecture. (b) UCF-Net-RBF-1.

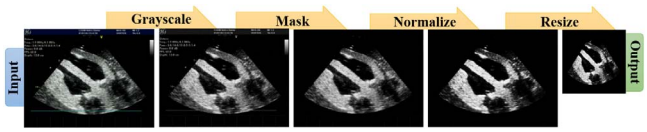


Fig. 3. Preprocessing the input B-mode ultrasound image stages.

followed by ReLU activation functions for the first two layers and linear activation function for the last layer. Fifty percent of dropouts are also considered active during training for the fully connected layers. The network returns $S \in \mathbb{R}^6$. S is considered as the DUIF which is used for adaptive compensator and external control loop design.

B. Data Collection and Training Procedure

UCF-Net is trained using pairs of input–output data consisting of ultrasound images and DUIF. DUIF corresponds to the poses of the ultrasound probe relative to a fixed frame attached to the base of the patient's body $\{\mathbf{B}\}$. To this end, an accurate photogrammetry sensor is used to detect $\{\mathbf{R}\}$ with respect to $\{\mathbf{B}\}$ and deriving ${}_R^B\mathbf{T}$. The pose of the ultrasound probe with respect to $\{\mathbf{R}\}$, i.e., ${}_T^R\mathbf{T}$ is achieved from robotic arm forward kinematics. Thus, the ultrasound probe's pose with respect to $\{\mathbf{B}\}$, can be extracted from ${}_T^B\mathbf{T} = {}_R^B\mathbf{T}\,{}_T^R\mathbf{T}$. To collect the required data, several tests with various parameters are conducted. While the robot is interacting with the phantom in random trajectories, the real-time ultrasound images are collected and labeled with the corresponding DUIF. Root mean square error (RMSE) is defined as the loss function, and ADAM optimizer is employed for training purposes.

Remark 1: It is important to note that different patients may have unique image features. For each patient, experts should define the main features of ultrasound images. Before training the UCF-Net, a calibration process is recommended to align these main features across patients, enhancing robustness and accuracy. However, this article omits this procedure, as all tests are conducted on a cardiac phantom.

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IV. Intelligent Adaptive Robust Force Controller

A. Adaptive Robust Force Control: Low-Level Controller

1) Orthogonal Sliding Manifold: The HPFC method takes advantage of orthogonality to define the sliding manifold σ as follows:

$$\sigma = \lambda_f \mathbf{Q} e_F + \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(e_X) \tag{17}$$

where $\lambda_f \in \mathbb{R}^+$, and $\lambda_p \in \mathbb{R}^+$ are constant coefficients. If $\sigma = 0$, the objective of the HPFC in (1) is achieved. Taking the time derivative of both sides of (2), (3), and (17), yields

$$\dot{e}_X = \dot{X} - \dot{X}_d = V - V_d, \dot{e}_F = \dot{F} - \dot{F}_d \tag{18}$$

$$\dot{\sigma} = \lambda_f \mathbf{Q} \dot{e}_F + \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(\dot{e}_X). \tag{19}$$

Substituting (14) and (18) into (19) yields

$$\dot{\sigma} = \lambda_f \mathbf{Q}\left({}^T_R\mathrm{f}\left(\widehat{k}_b V + \widehat{c}_b \dot{V}\right) + D\left(\widehat{k}_b \dot{d}_u + \widehat{c}_b \ddot{d}_u\right)\right) - D\lambda_f \dot{f}_{nd} + \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V) + \delta_L \tag{20}$$

where

$$\delta_L = \lambda_f D \dot{\tilde{\delta}}_b. \tag{21}$$

Since the velocity screw V is considered as the control input, and $\mathbf{Q}D = D$, (20) can be rewritten as follows:

$$\dot{\sigma} = u + \lambda_f D\left(\widehat{k}_b \dot{d}_u + \widehat{c}_b \ddot{d}_u - \dot{f}_{nd}\right) - \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V_d) + \delta_L \tag{22}$$

where u is the virtual control input to be defined using AFQSMC and has the following relation with V as follows:

$$\lambda_f \mathbf{Q}^T_R \mathrm{f}\left(\widehat{k}_b V + \widehat{c}_b \dot{V}\right) + \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V) = u. \tag{23}$$

The relation above can be reformulated as follows:

$$\lambda_f \mathbf{Q}^T_R \mathrm{f}\left(\widehat{k}_b V + \widehat{c}_b \dot{V}\right) + \lambda_p (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V) = \mathbf{Q}u + (\mathbf{I}-\mathbf{Q})u. \tag{24}$$

Due to the orthogonality principle, one may decompose (24) as follows:

$$(\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V) = \lambda_p^{-1}(\mathbf{I}-\mathbf{Q})u, \mathbf{Q}^T_R \mathrm{f}(V) = \int_0^t \widehat{c}_b^{-1} \mathbf{Q}\left(\lambda_f^{-1} u - \widehat{k}_b {}^T_R \mathrm{f}(V)\right) dt. \tag{25}$$

Thus, once u is defined, based on (20), one may derive V using the following relations:

$$V = {}^T_R\mathrm{f}^{-1}\left(\lambda_p^{-1}(\mathbf{I}-\mathbf{Q})u + \int_0^t \widehat{c}_b^{-1}\mathbf{Q}\left(\lambda_f^{-1}u - \widehat{k}_b {}^T_R\mathrm{f}(V)\right) dt\right). \tag{26}$$

2) AFQSMC: To force the orthogonal sliding manifold to converge to zero, AFQSMC is suggested, which not only is robust to existing uncertainties, but also filters the noisy feedback signals by exploiting a filtering mechanism within its control law. Thanks to its continuous control command, it also leads to a chattering-free control process. The adaptive uncertainty estimator/compensator based on UCF-RBF-Net-1 also contributes to higher tracking accuracy. Based on (22), AQFSMC is derived as follows:

$$u = \eta + \lambda_f D\left(\widehat{k}_b \dot{d}_u + \widehat{c}_b \ddot{d}_u - \dot{f}_{nd}\right) + (\mathbf{I}-\mathbf{Q})^T_R \mathrm{f}(V_d) - \widehat{L}_d \tag{27}$$

where

$$\begin{cases} \eta = -\mathbf{\Lambda_1^{-1}\Lambda_3}\sigma_C - \mathbf{\Lambda_1^{-1}}\mathrm{sat}\left(\boldsymbol{\phi}^{-1}\sigma_C\right) \\ \dot{\sigma}_C = -\mathbf{\Lambda_2}\sigma_C + \mathbf{\Lambda_1}\sigma \end{cases} \tag{28}$$

$$\mathrm{sat}\left(\boldsymbol{\phi}^{-1}\sigma_C\right) = \begin{cases} \boldsymbol{\phi}^{-1}\sigma_C, & \left|\boldsymbol{\phi}^{-1}\sigma_C\right| < K \\ K\ \ \mathrm{sign}\left(\boldsymbol{\phi}^{-1}\sigma_C\right), & \left|\boldsymbol{\phi}^{-1}\sigma_C\right| \geq K \end{cases}. \tag{29}$$

In the relations above, $\mathbf{\Lambda}_1 \in \mathbb{R}^{6\times6}$ to $\mathbf{\Lambda}_3 \in \mathbb{R}^{6\times6}$ are diagonal positive matrices of control design parameters; $\boldsymbol{\phi}^{-1} \in \mathbb{R}^{6\times6}$ is diagonal positive matrix of the boundary layer thickness; and $K \in \mathbb{R}^6$ is the saturation threshold whose value is defined based on the bounds of uncertainty of the system. $\widehat{L}_\mathrm{d} \in \mathbb{R}^6$ is the uncertainty compensator determined as follows:

$$\widehat{L}_d = \mathbf{W}H \tag{30}$$

where

$$H = \left[h_1, h_2, \ldots, h_{n_f}, 1\right]^\mathrm{T}, \mathbf{W} = \begin{bmatrix} w_{1,1}, w_{1,2}, \ldots, w_{1,n_f+1} \\ \vdots \\ w_{6,1}, w_{6,2}, \ldots, w_{6,n_f+1} \end{bmatrix} \tag{31}$$

where $h_i \in \mathbb{R}$ for $i \in \{1, \ldots n_f\}$ and $w_{k,l} \in \mathbb{R}$ for $k \in \{1, \ldots, 6\}$ are the nonlinear part's output and linear part's weights of the RBFNN, respectively [Fig. 2(b)]. h_i is calculated as follows:

$$h_i = \exp\left(-||P - C_i||^2 / 2G_i^2\right) \tag{32}$$

where $C_i \in \mathbb{R}^7$ and $G_i \in \mathbb{R}^+$, $i = 1, \ldots, n_f$ are the centers and variances of the RBFNNs, defined randomly and by the c-means clustering algorithm, respectively. UCF-Net-RBF-1 can also be separated into two parts: decoder parts (which is UCF-Net based on VGG-16) and RBFNN part. S is the vector of deep features plus $\widehat{f}_n$, i.e., $P = \begin{bmatrix} S^\mathrm{T} & f_n \end{bmatrix}^\mathrm{T}$ where $S \in \mathbb{R}^6$ is the vector of deep features returned by the UCF-Net, which should take unique values for different views over the patient's body. To this end, the pose of the ultrasound probe with respect to $\{\mathbf{B}\}$ is considered as S corresponding to the ultrasound image. Thus, the UCF-Net is trained on the set of pairs of the ultrasound image and corresponding ultrasound probe pose. Note that the hyperparameters of the UCF-Net are adjusted during offline training, and the parameters of the linear part of the RBFNNs are updated adaptively during control process using the following relation:

$$\dot{\widehat{\mathbf{W}}}^\mathrm{T} = 2r\left(\dot{\sigma}_c^\mathrm{T} + \omega\sigma_c^\mathrm{T}\right)\mathbf{\Lambda}_1 H^\mathrm{T} \tag{33}$$

where, $r \in \mathbb{R}$ is the adaptation rate and $\omega \in \mathbb{R}$ is one of the control design parameters to be defined. The stability proof of the AFQSMC is presented in Theorem 1.

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Assumption 1: The following relation holds for δ_L and $\bar{L}_d$:

$$|\delta_L| \le D_L | D_L \in \mathbb{R}^{6\times 1},\ D_L > 0,\ \left|\tilde{L}_d\right| \le \tilde{D}_L \,|\, \tilde{D}_L \in \mathbb{R}^{6\times 1},\ \tilde{D}_L > 0 \tag{34}$$

where $\tilde{L}_d \in \mathbb{R}^6$ is the exact solution error defined in (41).

Theorem 1: Under the condition of (35), applying the AFQSMC law of (27) to the dynamic system (22) leads to its stability with a quasi-sliding motion, i.e., $|\sigma| < \epsilon_\sigma$ for all $t \ge T_r$ where $\epsilon_\sigma > 0$ and $T_r > 0$ [20], provided that Assumption 1 is satisfied.

$$\begin{aligned} &\mathbf{\Lambda_1}, \mathbf{\Lambda_2}, \mathbf{\Lambda_3} > 0, \mathbf{K} > \mathbf{\Lambda_1}\tilde{D}_L, (\mathbf{\Lambda_2} - \omega\mathbf{I}) > \mathbf{0}, \\ &\sqrt{2}(\mathbf{\Lambda_2} - \omega\mathbf{I})^{\frac{1}{2}}\sqrt{2\omega}\mathbf{\Lambda_3}^{\frac{1}{2}} \ge \left|(\gamma\mathbf{I} - \omega\mathbf{\Lambda_2} - \mathbf{\Lambda_3})\right| + \mathbf{2\phi}, \\ &\mathbf{\Lambda_2^2} - \mathbf{4}\left(\mathbf{\Lambda_3} + \boldsymbol{\phi}^{-1}\right) \ge 0, 0 < \omega < 1, \gamma > 1. \end{aligned} \tag{35}$$

Proof. Substituting (27) into (22) yields

$$\begin{cases} \dot{\sigma} = -\mathbf{\Lambda_1^{-1}\Lambda_3}\sigma_C - \mathbf{\Lambda_1^{-1}}\mathrm{sat}\left(\boldsymbol{\phi}^{-1}\sigma_C\right) + \delta_L - \widehat{L}_d \\ \dot{\sigma}_C = -\mathbf{\Lambda_2}\sigma_C + \mathbf{\Lambda_1}\sigma \end{cases}. \tag{36}$$

By taking the time derivative of the second row of (36) and substituting into its first row, the following relation is obtained:

$$\ddot{\sigma}_C = -\mathbf{\Lambda_2}\dot{\sigma}_C - \mathbf{\Lambda_3}\sigma_C - \mathrm{sat}\left(\boldsymbol{\phi}^{-1}\sigma_C\right) + \mathbf{\Lambda_1}\left(\delta_L - \widehat{L}_d\right). \tag{37}$$

Consider the following candidate Lyapunov function:

$$V_L = \gamma\sigma_C^{\mathrm{T}}\sigma_C + 2\omega\sigma_C^{\mathrm{T}}\dot{\sigma}_C + \dot{\sigma}_C^{\mathrm{T}}\dot{\sigma}_C + \frac{1}{2r}\mathrm{tr}\left\{\bar{\mathbf{W}}\bar{\mathbf{W}}^{\mathrm{T}}\right\} \tag{38}$$

where $\bar{\mathbf{W}}$, the error between the estimated weight matrix $\widehat{\mathbf{W}}$ and optimal weight matrix $\mathbf{W}_\star$, is calculated as follows:

$$\bar{\mathbf{W}} = \mathbf{W}_\star - \widehat{\mathbf{W}}. \tag{39}$$

Note that $\mathbf{W}_\star$ and $\widehat{\mathbf{W}}$ meet the following relations:

$$\begin{aligned} &\widehat{L}_d = \widehat{\mathbf{W}}H,\ \ L_\star = \mathbf{W}_\star H \to \bar{L}_d = \bar{\mathbf{W}}H, \\ &\mathbf{W}_\star = \mathrm{Arg}\min_{\mathbf{W}}\left[\sup_{(\mathbf{I}_{US}, f_n)}\left|\widehat{\mathrm{L}}_d - \delta_L\right|\right]. \end{aligned} \tag{40}$$

Thus, one may define the exact solution error $\tilde{L}_d$ and the estimated solution error $\bar{L}_d$ as follows:

$$\tilde{L}_d = \delta_L - L_\star,\ \ \bar{L}_d = L_\star - \widehat{L}_d \to \delta_L - \widehat{L}_d = \tilde{L}_d + \bar{L}_d. \tag{41}$$

Based on (35), it can be deduced that $\omega < \gamma^{1/2}$. This inequality results in the following relation:

$$\left|2\omega\sigma_C^{\mathrm{T}}\dot{\sigma}_C\right| \le \left|2\gamma^{\frac{1}{2}}\sigma_C^{\mathrm{T}}\dot{\sigma}_C\right|,\ \ \left|\gamma\sigma_C^{\mathrm{T}}\sigma_C\right| \ge 0,\ \ \left|\dot{\sigma}_C^{\mathrm{T}}\dot{\sigma}_C\right| \ge 0. \tag{42}$$

Additionally, the following relation holds:

$$\left(\gamma^{\frac{1}{2}}\sigma_c + \dot{\sigma}_c\right)^{\mathrm{T}}\left(\gamma^{\frac{1}{2}}\sigma_c + \dot{\sigma}_c\right) = \gamma\sigma_C^{\mathrm{T}}\sigma_C + 2\gamma^{\frac{1}{2}}\sigma_C^{\mathrm{T}}\dot{\sigma}_C + \dot{\sigma}_C^{\mathrm{T}}\dot{\sigma}_C \ge 0. \tag{43}$$

Comparing (42) with (43) shows $V_L = 0$ when $(\sigma_C, \dot{\sigma}_C, \bar{\mathbf{W}}) = (0,0,0)$ and $V_L > 0$ when $(\sigma_C, \dot{\sigma}_C, \bar{\mathbf{W}}) \ne (0,0)$: the required condition of a Lyapunov function. $\dot{V}_L$ is obtained as follows:

$$\dot{V}_L = 2\gamma\sigma_C^{\mathrm{T}}\dot{\sigma}_C + 2\omega\dot{\sigma}_C^{\mathrm{T}}\dot{\sigma}_C + 2\omega\sigma_C^{\mathrm{T}}\ddot{\sigma}_C + 2\dot{\sigma}_C^{\mathrm{T}}\ddot{\sigma}_C + \frac{1}{r}\mathrm{tr}\left\{\bar{\mathbf{W}}\dot{\bar{\mathbf{W}}}^{\mathrm{T}}\right\}. \tag{44}$$

Substituting (40) and (41) into (44) yields

$$\begin{aligned} \dot{V}_L = {} & \dot{\sigma}_c^{\mathrm{T}}\left(2(\gamma\mathbf{I} - \omega\mathbf{\Lambda_2} - \mathbf{\Lambda_3})\sigma_c - 2\left(K\mathrm{sign}(\sigma_C)\right)\right) \\ & - 2\dot{\sigma}_c^{\mathrm{T}}(\mathbf{\Lambda_2} - \omega\mathbf{I})\dot{\sigma}_C - 2\omega\sigma_c^{\mathrm{T}}\mathbf{\Lambda_3}\sigma_C - 2\omega\sigma_c^{\mathrm{T}}\left(K\mathrm{sign}(\sigma_C)\right) \\ & + 2\left(\dot{\sigma}_c^{\mathrm{T}} + \omega\sigma_c^{\mathrm{T}}\right)\mathbf{\Lambda_1}\left(\tilde{L}_d + \bar{\mathbf{W}}H\right) + \frac{1}{r}\mathrm{tr}\left\{\bar{\mathbf{W}}\dot{\bar{\mathbf{W}}}^{\mathrm{T}}\right\}. \end{aligned} \tag{45}$$

Substituting (33) into (45) yields

$$\begin{aligned} \dot{V}_L = {} & \dot{\sigma}_c^{\mathrm{T}}\left(2(\gamma\mathbf{I} - \omega\mathbf{\Lambda_2} - \mathbf{\Lambda_3})\sigma_c - 2\left(K\mathrm{sign}(\sigma_C)\right)\right) \\ & - 2\dot{\sigma}_c^{\mathrm{T}}(\mathbf{\Lambda_2} - \omega\mathbf{I})\dot{\sigma}_C - 2\omega\sigma_c^{\mathrm{T}}\mathbf{\Lambda_3}\sigma_C - 2\omega\sigma_c^{\mathrm{T}}\left(K\mathrm{sign}(\sigma_C)\right) \\ & + 2\left(\dot{\sigma}_c^{\mathrm{T}} + \omega\sigma_c^{\mathrm{T}}\right)\mathbf{\Lambda_1}\tilde{L}_d. \end{aligned} \tag{46}$$

Based on Assumption 1 and (35), the following relations yield:

$$-2\omega\sigma_c^{\mathrm{T}}\left(K\mathrm{sign}(\sigma_C) - \mathbf{\Lambda_1}\tilde{L}_d\right) \le 0 \tag{47}$$

$$\begin{aligned} &\sqrt{2}(\mathbf{\Lambda_2} - \omega\mathbf{I})^{\frac{1}{2}}\sqrt{2\omega}\mathbf{\Lambda_3}^{\frac{1}{2}} \ge \left|(\gamma\mathbf{I} - \omega\mathbf{\Lambda_2} - \mathbf{\Lambda_3})\right| + 2\boldsymbol{\phi} \\ &\ge \left|\left((\gamma\mathbf{I} - \omega\mathbf{\Lambda_2} - \mathbf{\Lambda_3}) - \left(\frac{\mathbf{K}}{|\sigma_C|} - \frac{\mathbf{\Lambda_1}\tilde{L}_d}{|\sigma_C|}\right)\right)\right|. \end{aligned} \tag{48}$$

Comparing (48) and (47) with (45) yields

$$\dot{V}_L < -2\left((\mathbf{\Lambda_2} - \omega\mathbf{I})^{\frac{1}{2}}\dot{\sigma}_c + \sqrt{\omega}\mathbf{\Lambda_3}^{\frac{1}{2}}\sigma_c\right)^{\mathrm{T}}\left((\mathbf{\Lambda_2} - \omega\mathbf{I})^{\frac{1}{2}}\dot{\sigma}_c + \sqrt{\omega}\mathbf{\Lambda_3}^{\frac{1}{2}}\sigma_c\right). \tag{49}$$

Accordingly, when $\left|\phi^{-1}\sigma_C\right| \ge K$, one has $\dot{V}_L < 0$.

When $\left|\phi^{-1}\sigma_C\right| < K$, (37) is obtained as follows:

$$\ddot{\sigma}_C = -\mathbf{\Lambda_2}\dot{\sigma}_C - (\mathbf{\Lambda_3} + \boldsymbol{\phi}^{-1})\sigma_C + \mathbf{\Lambda_1}(\tilde{L}_d + \bar{L}_d). \tag{50}$$

The adaptive compensator contributes to leading $\bar{L}_d$ toward zero, i.e., $\lim_{n\to\infty}\bar{L}_d = 0$. Thus, since $\mathbf{\Lambda_1}$ to $\mathbf{\Lambda_3}$ and $\mathbf{\Phi}$ are constants, the relation above can be transformed to the frequency domain using Laplace transformation as follows:

$$\Sigma_c p^2 + \mathbf{\Lambda_2}\Sigma_c p + \left(\mathbf{\Lambda_3} + \boldsymbol{\phi}^{-1}\right)\Sigma_c = \mathbf{\Lambda_1}\tilde{\Delta}_{\mathrm{L}} \tag{51}$$

where p is Laplace operator, $\Sigma_{\mathrm{c}} = \mathcal{L}\{\sigma_c\}$ and $\tilde{\Delta}_{\mathrm{L}} = \mathcal{L}\{\tilde{L}_d\}$, where $\mathcal{L}\{\star\}$ is Laplace function. Thus, the transfer function $T_1(p) = \Sigma_c\tilde{\Delta}_{\mathrm{L}}^{-1}$ is obtained as follows:

$$\begin{aligned} &\mathrm{T}_1(p) = \Sigma_c\tilde{\Delta}_{\mathrm{L}}^{-1} = \mathbf{\Lambda_1}((p\mathbf{I} + \mathbf{R_1})(p\mathbf{I} + \mathbf{R_2}))^{-1}, \\ &\mathbf{R_1} = \frac{\mathbf{1}}{\mathbf{2}}\left(-\mathbf{\Lambda_2} + \left(\mathbf{\Lambda_2^2} - \mathbf{4}\left(\mathbf{\Lambda_3} + \boldsymbol{\phi}^{-1}\right)\right)^{\mathbf{1/2}}\right), \\ &\mathbf{R_2} = \frac{\mathbf{1}}{\mathbf{2}}\left(-\mathbf{\Lambda_2} - \left(\mathbf{\Lambda_2^2} - \mathbf{4}\left(\mathbf{\Lambda_3} + \boldsymbol{\phi}^{-1}\right)\right)^{\mathbf{1/2}}\right). \end{aligned} \tag{52}$$

Based on (35) and (52) has real poles. Also, taking Laplace of the second row of (36) results in the following transfer function:

$$\mathrm{T}_2(p) = \Sigma_c\Sigma^{-1} = (p\mathbf{I} + \mathbf{\Lambda_2})^{-1}\mathbf{\Lambda_1} \tag{53}$$

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where $\Sigma = L\{\sigma\}$. Thus, $T_3(p) = \Sigma\tilde{\Delta}_{\mathrm{L}}^{-1}$ can be obtained as follows:

$$\mathrm{T}_3(p) = \mathrm{T}_1(p)\mathrm{T}_2^{-1}(p) = \mathbf{\Lambda_1}\big((p\mathbf{I}+\mathbf{\Lambda_2})(p\mathbf{I}+\mathbf{R_1})(p\mathbf{I}+\mathbf{R_2})\big)^{-1}. \tag{54}$$

Based on (35), the numerator's coefficients of $T_3(p)$ are positive and real. Hence, it is stable. So, the response has its maximum magnitude in zero frequency [14]. It means that the response is bounded as $|\sigma| < \epsilon_\sigma$ where $\epsilon_\sigma = T_3(0) = (\mathbf{R_1})^{-1}(\mathbf{R_2})^{-1}\mathbf{\Lambda_1}(\mathbf{\Lambda_2})^{-1} = \mathbf{\Lambda_1^2}\big(\mathbf{\Lambda_3}+\boldsymbol{\phi}^{-\mathbf{1}}\big)^{-1}\mathbf{\Lambda_2^{-1}}$. Thus, AFQSMC is a quasi-SMC, and the theorem is proved.

Remark 2: $T_2(p)$ in (53) is a low-pass filter that exists in the reaching law in (28). Thus, it filters out the existing noises in the sliding manifold σ emanated from force sensor and camera measurement noises. Also, AFQSMC exploits a continuous function in the control law which makes it chattering free.

Remark 3: Based (41), it can be deduced that $\tilde{D}_L \ll D_L$. Thus, $\boldsymbol{K}$ can be selected smaller leading to a smaller boundary layer and, higher tracking accuracy, and less chattering.

B. External Control Loop: High-Level Controller

Once an adaptive robust controller is developed for interaction force control, the primary concern becomes determining the desired interaction force, which depends on the ultrasound image quality and the patient's safety and comfort. On the one hand, if the interaction force applied during cardiac ultrasound scanning is inadequate, the ultrasound probe may not fully contact the skin, leading to low-quality image acquisition. However, excessive interaction force can cause discomfort to the patient or result in injury. Thus, in this study, by exploiting the uniqueness of the features of the UCF-Net, a controller is developed to define the desired force based on the ultrasound image feedback. A new model based on the UCF-Net, followed by an RBFNN, is developed, named UCF-Net-RBF-2, which processes the ultrasound image and returns the desired value.

1) USCF-Net-RBF-2: Experiments show that the minimum requirement of the applied force mostly depends on the pose of the ultrasound probe on the patient's body, e.g., the required force for echo scanning of the personal short axis (PSAX) view is less than that for SC. To solve this issue, the deep features extracted by the UCF-Net are feedforward to an RBFNN, leading to UCF-Net-RBF-2, to define the desired force as follows:

$$f_{nd}(S) = \sum_{i=1}^{4}\left(\bar{f}_{ndi}\bar{h}_i + \bar{f}_{ndm}\bar{h}_m\right) \Bigg/ \left(\sum_{i=1}^{4}\bar{h}_i + \bar{h}_m\right) \tag{55}$$

where

$$\bar{h}_i = \exp\left(\frac{\|S-\bar{C}_i\|^2}{2\bar{G}_i^2}\right), \quad \bar{h}_m = 1 - \max\{\bar{h}_1, \bar{h}_2, \bar{h}_3, \bar{h}_4\}. \tag{56}$$

The RBFs centers $\bar{C}_i \in \mathbb{R}^6$, $i = 1, \ldots, 4$ are features of the main views. $\bar{G}_i^2 \in \mathbb{R}^+$, $i = 1, \ldots, 4$ are the RBF's radii. $\bar{f}_{ndm}$ is the minimum force required for interacting [Fig. 4(b)]. The architecture of UCF-Net-RBF-2 is demonstrated in Fig. 2(b). To collect the required data, first, the robot goes to the four main cardiac views confirmed by the cardiologists (in this study, all the tests are taken on a cardiac phantom training block). Then, the optimal interaction force reference values are defined by the cardiologist should be the minimum force that gives the best ultrasound image of the intended view. This procedure is repeated four times around each view.

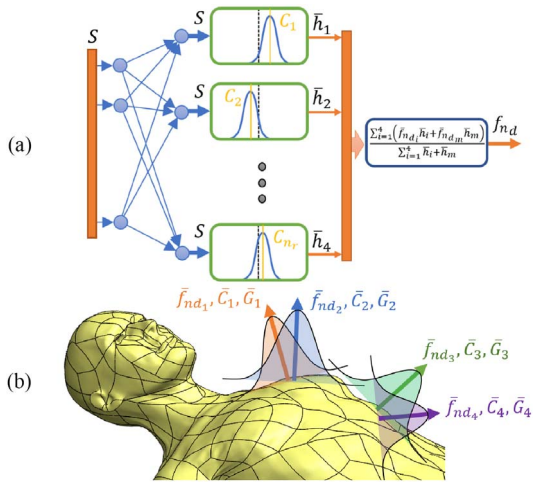


Fig. 4. (a) UCF-Net-RBF-2. (b) RBF parameters of UCF-Net-RBF-2.

Remark 4: The desired interaction force may vary from one patient to another. For example, a person with thicker fat tissues may require more interaction force. To address this issue, the last layer of the encoder part of UCF-Net can have an additional output channel as the correction factor, which corrects the determined desired force. However, in this study, the data were collected from a cardiac phantom. Thus, the correction factor is considered as one, which is not presented in the figures and formulas.

2) External Control Loop Design: The external control loop determines the desired force during the interaction. Accordingly, the ultrasound image is acquired and fed to the UCF-Net, whose output is the desired interaction force fed to the HPFC in the internal control loop. However, at the beginning of the interaction, adequate interaction force is required to get US-image clearly to feed the UCF-Net. To this end, an algorithm is developed that once the ultrasound probe touches the skin, ultrasound image acquisition starts, and the integration force increases till a clear ultrasound image appears. This can be detected by the UCF-Net. Once the UCF-Net returns the feature values within the acceptable range, the quality of the ultrasound image is good enough to be used in the external control loop. Then, the ultrasound image is fed to the UCF-Net, and the external control loop is activated. The algorithm is also responsible for contact detection and switching between free motion (when the ultrasound probe is not in contact with the patient's body) and HPFC. This procedure is presented in Algorithm 1. The flowchart of the algorithm is also presented in Fig. 5. The block diagram of the Algorithm 1 is presented in Fig. 6.

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Algorithm 1: Physical Interaction Detection Algorithm.

Set ($k = 0$, ϵ_l, and ϵ_h); ϵ_l and ϵ_H are the low and high value of force thresholds;
Set ($inx_c = 0$, $inx_u = 0$); % inx_c and inx_u are contact and image quality indicators;
Set ($f_{n_d} = 5$, $\Delta f_n = 0.1$ N);
while (True)
  Measure the interaction force $f_d(t_k)$;
  IF (($f_d(t_k) > \epsilon_h$) or ($f_d(t_k) \geq \epsilon_l$ and $inx_C = 1$))
    $inx_C = 1$;
  End
  IF ($f_d(t_k) < \epsilon_l$)
    $inx_C = 0$;
  End
  IF (inx_C==1)
    Capture US image and extract the image features (S_k) using USCF-Net;
    IF (S_k is in the expected range)
      $inx_u = 1$;
    Else
      $inx_u = 0$;
    End
    IF (inx_u == 1)
      Activate external control loop;
    Else
      Deactivate external control loop;
    End
    IF (inx_C == 1 and inx_u == 0 and $f_{nd} < 20$)
      Increase the desired force f_{n_d} by Δf_n;
    End
  End
  $k = k + 1$;
End

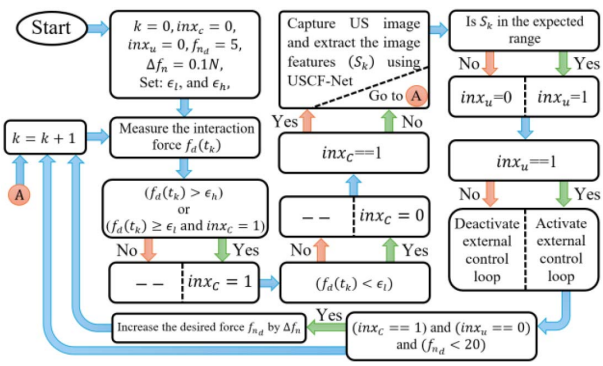


Fig. 5. Flowchart of the physical interaction detection algorithm.

V. Experimental Results and Discussion

A. Experimental Setup

The CURS setup includes a Universal Robot UR5e equipped with a built-in torque/force sensor, and a phased-array ultrasound probe, the GE-M4S with 20 × 28 mm footprint (5.6e–4m^2), connected to a Vivid-q ultrasound machine. A passive prismatic joint has been manufactured to hold both the ultrasound probe and a camera. A CAE cardiac Blue Phantom is included in the setup to mimic the human body for heart sonography. An Intel RealSense camera D405 is attached to the end-effector, assisting the remote sonographer in maneuvering the robot to the intended view using a Logitech joystick. All devices and sensors are integrated with the robot operating system (ROS) Noetic version. The system is powered by a PC that is equipped with an Intel Core i9-13900 CPU, 64gb RAM DDR5-4800, and a Nvidia RTX 4090 GPU. The sampling times of the inner and outer control loops are 5 and 40 ms, respectively. The experimental setup is depicted in Fig. 7.

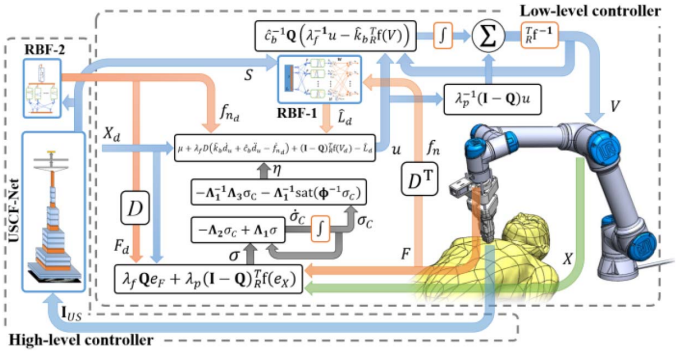


Fig. 6. Schematic of the proposed controller.

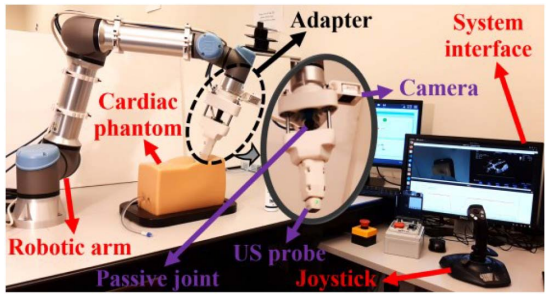


Fig. 7. Experimental setup.

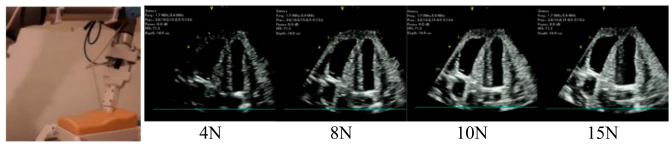


Fig. 8. (a) CURS interacting with the phantom at A4C. (b) Ultrasound images of the A4C view for different interaction force values.

B. Training UCF-Net

Fig. 8 shows the CURS interacting with the phantom at A4C view for the first test. It shows the effect of the value of the interaction force on the image quality and how the optimal value is chosen. Referring to Fig. 8(b), the image reaches the highest quality at 15 N (8.4e–3Pa) and does not increase afterward. Thus, 15 N is selected for this view. The same procedure is performed for other main views, including PSAX, parasternal long axis view (PLAX), SC, and A4C views which are established by a cardiologist. The desired pressure is also calculated considering

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TABLE I
PARAMETERS OF RBF PART OF THE UCF-NET-RBF-2

View	PSAX	PLAX	SC	A4C
Force/ pressure value	10 N/5.6e-3Pa	16 N/8.9e-3Pa	27 N/15.1e-3Pa	15 N/8.4e-3Pa
radius	2	5	3	8
Center (main image feature)	$\begin{bmatrix}0.4742\\0.3696\\0.4316\\0.5649\\0.4028\\0.0509\end{bmatrix}$	$\begin{bmatrix}0.1222\\0.4643\\0.2104\\0.5391\\0.5773\\1.5361\end{bmatrix}$	$\begin{bmatrix}0.8205\\0.3202\\0.4054\\0.6886\\0.1094\\0.7848\end{bmatrix}$	$\begin{bmatrix}0.8216\\0.6521\\0.4577\\0.4487\\0.1210\\0.4749\end{bmatrix}$

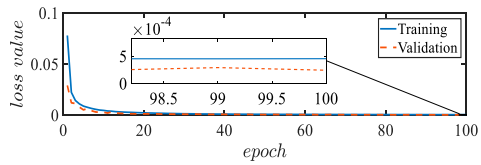


Fig. 9. Training process: loss value as a function of the numbers of epochs.

the probe footprint. The interaction force and pressure for the main views and their corresponding RBF centers and radii are measured and tabulated in Table I. The experiments show that the maximum required interaction force/pressure is 27 N/15.1e–3Pa (for the A4C view).

The UCF-Net is trained on a data set containing 10 152 ultrasound images and corresponding image features. The ADAMS optimizer is employed for training the network with an initial learning rate of 1.0e–4 for 100 epochs and mini-batch size of 200. Overall, 70% of the data is allocated for training (7107 images) and 20% (2031 images) for validation, which are selected randomly. The rest of the data is used for testing (1014). The training process results are presented in Fig. 9. The results show that the validation loss has been smaller than the training loss while close to it, indicating that the model is trained well. The training process takes almost 19 min. Also, the computation time for the trained model to take a B-mode ultrasound image and return the DUIF is 33 ms.

C. Experiment Design

Test #1 is designed to demonstrate the effectiveness of the proposed AFQSMC for force control in the internal loop. To this end, the robot moves to the PSAV by the expert sonographer and starts the interaction for two different cases: regulation on 12 N (Case #1) and tracking of the desired signal $f_{nd} = 11 + 4\cos(0.4t + 2.2)N$ (Case#2). Besides, well-known methods such as P/PI, SMC, and FQSMC are employed for a comparative study focusing on control performance. To have a fair comparison, the optimization method presented in [15] is employed to tune the control gains. The control gains of the proposed method are: $\lambda_f = 0.126$; $\lambda_p = 1$; $\mathbf{\Lambda_1} = 10\mathrm{I}_{6\times6}$; $\mathbf{\Lambda_2} = 10\mathrm{I}_{6\times6}$; $\mathbf{\Lambda_3} = \begin{bmatrix}1.912\mathrm{I}_{3\times3} & 0_{3\times3}\\ 0_{3\times3} & 12.854\mathrm{I}_{3\times3}\end{bmatrix}$; $\boldsymbol{\phi} = \begin{bmatrix}1.953\mathrm{I}_{3\times3} & 0_{3\times3}\\ 0_{3\times3} & 0.425\mathrm{I}_{3\times3}\end{bmatrix}$; $K = \begin{bmatrix}0.3\mathrm{I}_{3\times3} & 0_{3\times3}\\ 0_{3\times3} & 1\mathrm{I}_{3\times3}\end{bmatrix}$. Additionally, to investigate the effect of incorporating the passive joint on the performance of the interaction, another test is conducted on case #1, both with and without the passive joint (with the prismatic joint locked so it cannot move), using AFQSMC.

In *Test #2*, the performance of the external loop and its effect on the ultrasound image quality is evaluated and compared to the scenario where the high-level controller is deactivated. For this purpose, the robot begins interacting with the phantom at the PSA view and sequentially moves to the next view. This procedure is repeated until all views have been covered.

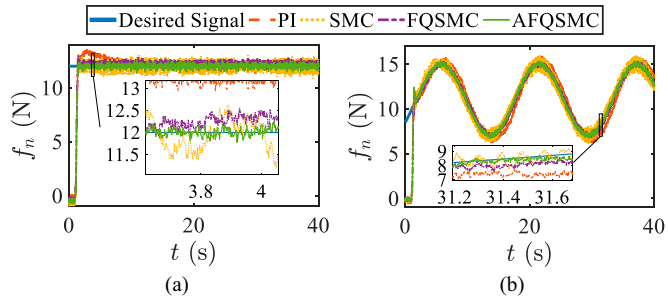


Fig. 10. Interaction force control. (a) case#1. (b) case#2.

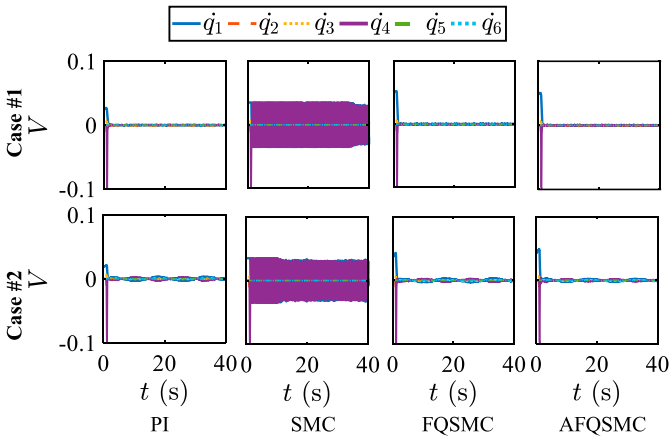


Fig. 11. Control input for the first test.

D. Experiment Results

The plots of interaction force control results for test #1 are shown in Fig. 10. Referring to Fig. 10, AFQSMC has tracked the desired force signal with a higher convergence rate and accuracy for both cases. SMC has high accuracy, although it suffers from significant chattering. FQSMC has a higher accuracy than PI but less than AFQSMC and SMC while chattering-free. The control input of the system for all methods is plotted in Fig. 11, showing the effectiveness of the proposed method in reducing chattering. The reason is that the boundary layer of FQSMC leads to lower tracking accuracy in favor of chattering reduction, while the adaptive part of the AFQSMC contributes to estimating the uncertainties and compensating for them in the controller, resulting in a lower boundary layer thickness and higher

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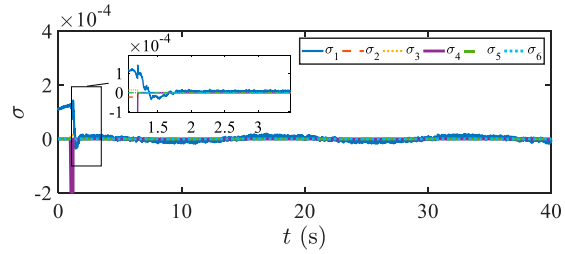


Fig. 12. Sliding variable of AFQSMC for test #1 – C#2.

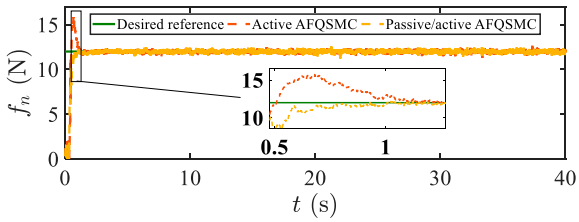


Fig. 13. Performance comparison of the system's fast initial impact with and without the passive joint.

TABLE II
COMPUTED INDICATORS FOR INTERACTION FORCE CONTROL OF TEST #1 [MM]

	Case #1				Case #2			
	P/PI	SMC	FQSMC	AFQSMC	P/PI	SMC	FQSMC	AFQSMC
RMS	0.1752	0.3706	0.2873	0.1468	0.9043	0.3758	0.3988	0.2439
STD	0.1728	0.3705	0.1468	0.1460	0.9027	0.3758	0.3981	0.2436
Max	1.6329	1.1592	0.8506	0.7130	1.8915	1.3503	0.9913	0.7430

accuracy. The results are the same as those for tracking the desired signal in case#2. Additionally, the sliding variable signals for AFQSMCs (Test #2 Case #2) are plotted in Fig. 12 to investigate the achievement of the sliding phase. The plot shows that the sliding surface has converged to zero and is maintained around it within the boundary layer, indicating that the sliding mode is achieved.

The results of comparing the initial impact with and without the passive joint are presented in Fig. 13. Referring to this figure, there is an intense spike of interaction force up to 15.8 N, while there is no overshoot for the case that includes the passive joint. This demonstrates the effectiveness of the passive joint in improving control performance and safety.

To have a quantitative comparison, numerical indicators such as root mean square (RMS), standard deviation (STD), and Max are computed for the control results and tabulated in Table II. According to Table II, AFQSMC is superior to other methods in terms of all indicators. The improvement percentages are illustrated in Fig. 14. AFQSMC has improved the system's performance by at least 0.5% compared to FQSMC in terms of STD for the regulation case. Note that FQSMC is chattering-free. Thus, AFQSMC has minimum improvement for STD that indicates chattering amount. The maximum improvement is 73.3% compared to PI in terms of RMS for the tracking case, indicating tracking accuracy. This improvement is maximum since although the integrator part in the PI controller improves the settling error for regulation case, it deteriorates the tracking performance.

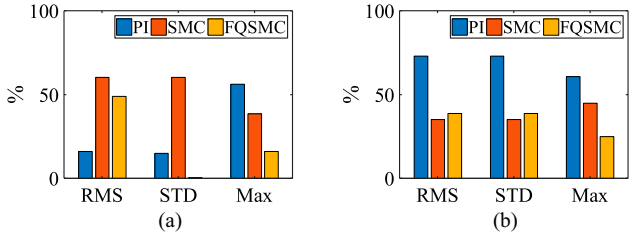


Fig. 14. Interaction force error improvement by AFQSMC. (a) case #1. (b) case #2.



Fig. 15. Scanning main cardiac views by the CURS.

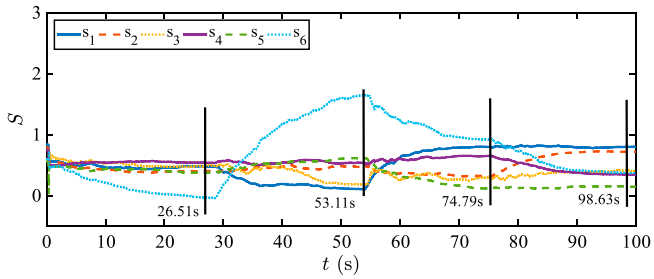


Fig. 16. DUIF during the interaction with the phantom on the predefined trajectory passing through the main cardiac views.

Overall, the average improvement compared to PI, SMC, and FQSMC are 27.2%, 53.08%, and 21.87%, respectively for case #1 and 68.25%, 38.42%, and 34.23%, respectively, for case #2. These improvement percentages confirm the superiority of the AFQSMC over other methods.

The sequence of scanning the main cardiac views by the CURS is demonstrated in Fig. 15. The generated deep features for this test are plotted in Fig. 16. Referring to this figure, once the features are close to the predefined features, the generated desired force becomes close to the predefined desired force presented in Table I. The results of force control and acquired ultrasound images on the predefined trajectory for both cases are presented in Fig. 17. According to this figure, the external control loop has generated the desired interaction force value, which reaches its predefined value when the ultrasound probe is at the determined view. The results show that when the external control loop is activated, the ultrasound image quality in determined cardiac views is as expected and high enough to acquire high-quality ultrasound images. This is not true for case #1 where the interaction force is considered constant. Although the results show the superiority of the proposed method and its feasibility for practical applications, one limitation is the computational burden required by the AI model, which needs high-efficient and powerful computer station, such as the one used in this study. This restricts the proposed approach from being implementable

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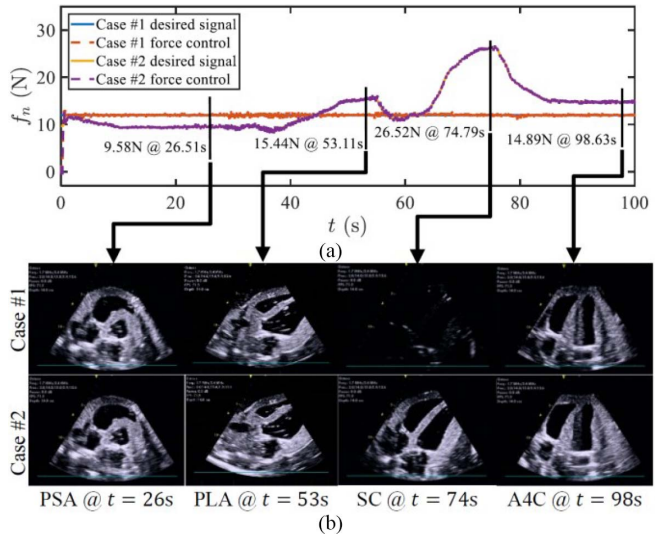


Fig. 17. (a) Interaction force. (b) Acquired ultrasound images during interaction.

on a normal personal computer. The video of T#2-C#2 is available on the following link: https://youtu.be/Z2iLaaQBFYY.

VI. Conclusion

This article proposed an AI-powered interaction force control method for CURS, encompassing both internal and external control loops. The internal loop was characterized using HPFC, incorporating OSM and AFSMC, while an AI model named UCF-Net was developed to process ultrasound images to extract six unique features. These features were utilized for adaptive compensation in the internal controller and for defining the desired interaction force value in the external controller. The experiments conducted on a cardiac phantom demonstrated the superiority of the proposed HPFC method in the internal loop over other well-known methods in terms of tracking accuracy, robustness against uncertainty, and chattering reduction. The external controller's effectiveness was also shown compared to the case where a constant desired force was used. Future work includes the development of a visual servoing method to automatically move toward the desired cardiac views, coupled with a path/task planning algorithm. Additionally, exploring different AI models with higher efficiency for UCF-Net is suggested to make it implementable on a wider range of computers or embedded systems. Exploring the effect of the communication time delay on the system performance is also an open topic.

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